

LAB NOTES: SQUID CHARACTERISATION

1. Junction V-I Characteristics

We initiated the measurement in the superconducting (SC) regime, observing the transition as the temperature increased towards the non-SC state.

- **Setup:** Drive current set to $120\mu\text{A}$ with a $10\text{k}\Omega$ bias resistor.
- **Energy gap:** The SC regime is governed by the energy gap $\Delta(T) \propto T_c \sqrt{1 - T/T_c}$, which represents the binding energy of Cooper pairs. The modulation of the critical current allows for the mapping of the density of states (DOS) threshold.
- **Params:** From the I-V curves (Fig. 1), the critical current was determined to be $I_c = 82\mu\text{A}$.

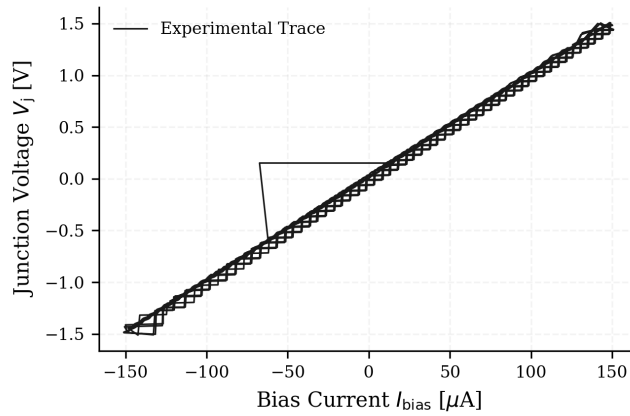


Fig 1: Single Josephson Junction I-V trace showing SC transition.

2. SQUID I-V Response

The flux bias current I_b was swept $\pm 300\mu\text{A}$ to tune the critical current of the SQUID loop. Adaptive gain was manually applied at the oscilloscope to maintain signal resolution across the modulation depth.

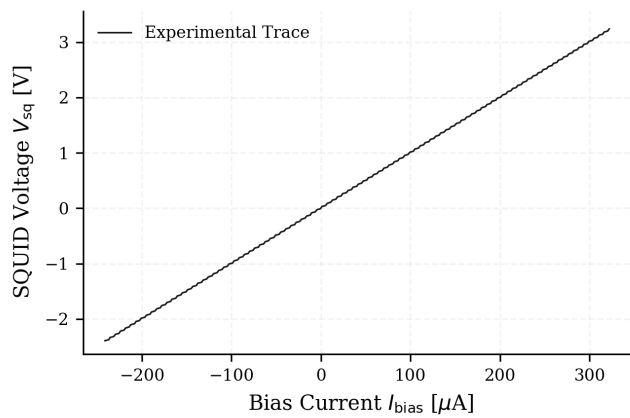


Fig 2: SQUID I-V characteristics under varying magnetic flux.

3. Flux response and Mutual Inductance

A 41 Hz triangle drive was used for flux biasing, resulting in an interference response of approximately 400(100) Hz, indicating the traversal of multiple flux quanta within a single sweep.

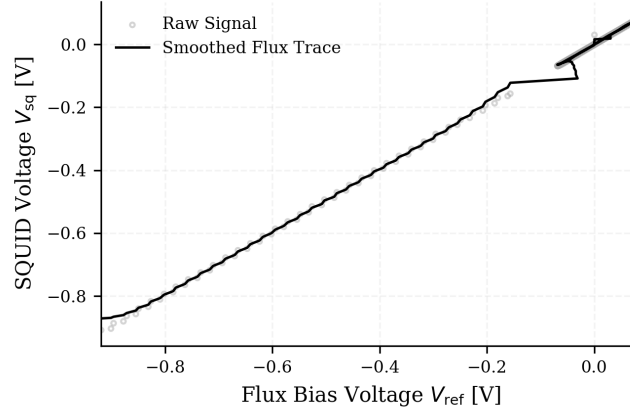


Fig 3: Interference pattern as a function of flux bias voltage V_{ref} .

Inductance Extraction (Task 3)

Automated peak detection on 14 analyzed periods yielded the following results:

Metric	Value
Bias Voltage Period ΔV_{ref}	0.8293 V
Bias Current Period ΔI_{flux}	82.93 μA
Mutual inductance M	24.94 nH

The extracted M is highly consistent with manual Gregorian-style estimation based on $\Delta I_{\text{flux}} = \Phi_0/M$.